2.68 2D Matrices: Eigenvalues and Eigenvectors

Worked Solutions

1 a)

$$\begin{pmatrix} 4 & 2 \\ 1 & 5 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \lambda \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 6 \\ -3 \end{pmatrix} = \lambda \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$\therefore \lambda = 3$$

1 b)

$$\begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 5 \\ 10 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\therefore \lambda = 5$$

1 c)

$$\begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -2 \\ 4 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\therefore \lambda = -2$$

2 a)

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 3 & -5 \\ 0 & -2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3-\lambda & -5 \\ 0 & -2-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (3-\lambda)(-2-\lambda) = \lambda^2 - \lambda - 6$$

$$\therefore \lambda^2 - \lambda - 6 = 0 \quad \text{as req.}$$

2 b)

$\lambda = 3, \lambda = -2$

3 a)

$$\begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4-\lambda & 2 \\ 1 & 3-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10$$

So $(\lambda - 2)(\lambda - 5) = 0$

$$\therefore \lambda = 2 \text{ or } \lambda = 5$$

3 b)

$$\begin{pmatrix} -3 & 6 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} -3 & 6 \\ -2 & 1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -3-\lambda & 6 \\ -2 & 1-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (-3-\lambda)(1-\lambda) - (-12) = \lambda^2 + 2\lambda + 9$$

So $\lambda^2 + 2\lambda + 9 = 0$

$$\therefore \lambda = -1 \pm 2\sqrt{2}i$$

3 c)

$$\begin{pmatrix} 0 & -5 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 0 & -5 \\ 1 & 0 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -\lambda & -5 \\ 1 & -\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 - 5$$

So $\lambda^2 - 5 = 0$

$$\therefore \lambda = \pm\sqrt{5}i$$

3 d)

$$\begin{pmatrix} 2 & 7 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 2 & 7 \\ 0 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2-\lambda & 7 \\ 0 & 2-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (2-\lambda)(2-\lambda) - 0$$

So $(2 - \lambda)(2 - \lambda) = 0$

$$\therefore \lambda = 2$$

3 e)

$$\begin{pmatrix} -1 & 4 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} -1 & 4 \\ -3 & 5 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1-\lambda & 4 \\ -3 & 5-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (-1-\lambda)(5-\lambda) - (-12) = \lambda^2 - 4\lambda + 7$$

So $\lambda^2 - 4\lambda + 7 = 0$

$$\therefore \lambda = 2 \pm \sqrt{3}i$$

3 f)

$$\begin{pmatrix} 6 & -1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 6 & -1 \\ 2 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 6-\lambda & -1 \\ 2 & 3-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (6-\lambda)(3-\lambda) - (-2) = \lambda^2 - 9\lambda + 20$$

So $(\lambda - 4)(\lambda - 5) = 0$

$$\therefore \lambda = 4 \text{ or } \lambda = 5$$

3 g)

$$\begin{pmatrix} \frac{1}{2} & -\frac{3}{4} \\ \frac{3}{4} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} \frac{1}{2} & -\frac{3}{4} \\ \frac{3}{4} & \frac{1}{2} \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}-\lambda & -\frac{3}{4} \\ \frac{3}{4} & \frac{1}{2}-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (\frac{1}{2} - \lambda)(\frac{1}{2} - \lambda) + \frac{9}{16} = \lambda^2 - \lambda + \frac{13}{16}$$

So $\lambda^2 - \lambda + \frac{13}{16} = 0 \Rightarrow 16\lambda^2 - 16\lambda + 13 = 0$

$$\therefore \lambda = \frac{1}{2} \pm \frac{3}{4}i$$

4 a)

$$\begin{pmatrix} 5 & 2 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 5 & 2 \\ 1 & 4 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 5-\lambda & 2 \\ 1 & 4-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (5-\lambda)(4-\lambda) - 2 = \lambda^2 - 9\lambda + 18$$

$$\therefore \lambda^2 - 9\lambda + 18 = 0 \quad \text{as req.}$$

4 b)

So $(\lambda - 3)(\lambda - 6) = 0$

$$\therefore \lambda = 3 \text{ or } \lambda = 6$$

When $\lambda_1 = 3$ then $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}$

$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow 2x + 2y = 0 \text{ so } -y = x \therefore \mathbf{v}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

When $\lambda_2 = 6$ then $\mathbf{A} - 6\mathbf{I} = \begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix}$

$$\begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -x + 2y = 0 \text{ so } 2y = x \therefore \mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\therefore \lambda_1 = 3, \mathbf{v}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \text{ and } \lambda_2 = 6, \mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

5 a)

$$\begin{pmatrix} 4 & 2 \\ -3 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 4 & 2 \\ -3 & -3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 - \lambda & 2 \\ -3 & -3 - \lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (4 - \lambda)(-3 - \lambda) + 6 = \lambda^2 - \lambda - 6$$

$$\text{So } (\lambda + 2)(\lambda - 3) = 0$$

$$\therefore \lambda = -2 \text{ or } \lambda = 3$$

When $\lambda_1 = -2$ then $\mathbf{A} + 2\mathbf{I} = \begin{pmatrix} 6 & 2 \\ -3 & -1 \end{pmatrix}$

$$\begin{pmatrix} 6 & 2 \\ -3 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow 6x + 2y = 0 \text{ so } y = -3x \therefore \mathbf{v}_1 = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

When $\lambda_2 = 3$ then $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} 1 & 2 \\ -3 & -6 \end{pmatrix}$

$$\begin{pmatrix} 1 & 2 \\ -3 & -6 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow x + 2y = 0 \text{ so } -2y = x \therefore \mathbf{v}_2 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$$\therefore \lambda_1 = -2, \mathbf{v}_1 = \begin{pmatrix} 1 \\ -3 \end{pmatrix} \text{ and } \lambda_2 = 3, \mathbf{v}_2 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

5 b)

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} -87 & 120 \\ -64 & 89 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -87-\lambda & 120 \\ -64 & 89-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (-87-\lambda)(89-\lambda) + 64 \times 120 = \lambda^2 - 2\lambda - 7743 + 7680$$

$$\Rightarrow \lambda^2 - 2\lambda - 63 = (\lambda - 9)(\lambda + 7) = 0$$

$$\therefore \lambda = 9 \text{ or } \lambda = -7$$

When $\lambda_1 = 9$ then $\mathbf{A} - 9\mathbf{I} = \begin{pmatrix} -96 & 120 \\ -64 & 80 \end{pmatrix}$

$$\begin{pmatrix} -96 & 120 \\ -64 & 80 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -96x + 120y = 0 \text{ so } 5y = 4x \therefore \mathbf{v}_1 = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$$

When $\lambda_2 = -7$ then $\mathbf{A} + 7\mathbf{I} = \begin{pmatrix} -80 & 120 \\ -64 & 96 \end{pmatrix}$

$$\begin{pmatrix} -80 & 120 \\ -64 & 96 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -80x + 120y = 0 \text{ so } 3y = 2x \therefore \mathbf{v}_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$\therefore \lambda_1 = 9, \mathbf{v}_1 = \begin{pmatrix} 5 \\ 4 \end{pmatrix} \text{ and } \lambda_2 = -7, \mathbf{v}_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

5 c)

$$\begin{pmatrix} 60 & 10 \\ -10 & 40 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 60 & 10 \\ -10 & 40 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 60-\lambda & 10 \\ -10 & 40-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (60-\lambda)(40-\lambda) + 100 = \lambda^2 - 100\lambda + 2500$$

$$\text{So } (\lambda - 50)^2 = 0$$

$$\therefore \lambda = 50$$

When $\lambda = 50$ then $\mathbf{A} - 50\mathbf{I} = \begin{pmatrix} 10 & 10 \\ -10 & -10 \end{pmatrix}$

$$\begin{pmatrix} 10 & 10 \\ -10 & -10 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow 10x + 10y = 0 \text{ so } y = -x \therefore \mathbf{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$\therefore \lambda = 50, \mathbf{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

5 d)

$$\begin{pmatrix} 67 & -54 \\ 27 & -32 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 67 & -54 \\ 27 & -32 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 67-\lambda & -54 \\ 27 & -32-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (67-\lambda)(-32-\lambda) - 54 \times 27 = \lambda^2 - 35\lambda - 686$$

$$\text{So } (\lambda - 49)(\lambda + 14) = 0$$

$$\therefore \lambda = 49 \text{ or } \lambda = -14$$

When $\lambda_1 = 49$ then $\mathbf{A} - 49\mathbf{I} = \begin{pmatrix} 18 & -54 \\ 27 & -81 \end{pmatrix}$

$$\begin{pmatrix} 18 & -54 \\ 27 & -81 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow 18x - 54y = 0 \text{ so } 3y = x \therefore \mathbf{v}_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

When $\lambda_2 = -14$ then $\mathbf{A} + 14\mathbf{I} = \begin{pmatrix} 81 & -54 \\ 27 & -18 \end{pmatrix}$

$$\begin{pmatrix} 81 & -54 \\ 27 & -18 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow 81x - 54y = 0 \text{ so } 2y = 3x \therefore \mathbf{v}_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$\therefore \lambda_1 = 49, \mathbf{v}_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \text{ and } \lambda_2 = -14, \mathbf{v}_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

5 e)

$$\begin{pmatrix} \frac{4}{15} & \frac{2}{3} \\ \frac{1}{3} & -\frac{1}{15} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} \frac{4}{15} & \frac{2}{3} \\ \frac{1}{3} & -\frac{1}{15} \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{4}{15}-\lambda & \frac{2}{3} \\ \frac{1}{3} & -\frac{1}{15}-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (\frac{4}{15} - \lambda)(-\frac{1}{15} - \lambda) - \frac{2}{9} = \lambda^2 - \frac{1}{5}\lambda - \frac{6}{25}$$

$$\Rightarrow 25\lambda^2 - 5\lambda - 6 = 0$$

$$\therefore \lambda = -\frac{2}{5} \text{ or } \lambda = \frac{3}{5}$$

When $\lambda_1 = -\frac{2}{5}$ then $\mathbf{A} + \frac{2}{5}\mathbf{I} = \begin{pmatrix} \frac{2}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix}$

$$\begin{pmatrix} \frac{2}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow \frac{2}{3}x + \frac{2}{3}y = 0 \text{ so } y = -x \therefore \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

When $\lambda_2 = \frac{3}{5}$ then $\mathbf{A} - \frac{3}{5}\mathbf{I} = \begin{pmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{pmatrix}$

$$\begin{pmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -\frac{1}{3}x + \frac{2}{3}y = 0 \text{ so } 2y = x \therefore \mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\therefore \lambda_1 = -\frac{2}{5}, \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ and } \lambda_2 = \frac{3}{5}, \mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

5 f)

$$\begin{pmatrix} -\frac{26}{15} & \frac{4}{5} \\ \frac{4}{5} & \frac{2}{5} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} -\frac{26}{15} & \frac{4}{5} \\ \frac{4}{5} & \frac{2}{5} \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -\frac{26}{15} - \lambda & \frac{4}{5} \\ \frac{4}{5} & \frac{2}{5} - \lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (-\frac{26}{15} - \lambda)(\frac{2}{5} - \lambda) - \frac{16}{25} = \lambda^2 + \frac{4}{3}\lambda - \frac{4}{3}$$

$$\Rightarrow 3\lambda^2 + 4\lambda - 4 = 0$$

$$\therefore \lambda = -2 \text{ or } \lambda = \frac{2}{3}$$

When $\lambda_1 = -2$ then $\mathbf{A} + 2\mathbf{I} = \begin{pmatrix} \frac{4}{15} & \frac{4}{5} \\ \frac{4}{5} & \frac{12}{5} \end{pmatrix}$

$$\begin{pmatrix} \frac{4}{15} & \frac{4}{5} \\ \frac{4}{5} & \frac{12}{5} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow \frac{4}{15}x + \frac{4}{5}y = 0 \text{ so } 3y = -x \therefore \mathbf{v}_1 = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

When $\lambda_2 = \frac{2}{3}$ then $\mathbf{A} - \frac{2}{3}\mathbf{I} = \begin{pmatrix} -\frac{12}{5} & \frac{4}{5} \\ \frac{4}{5} & -\frac{4}{15} \end{pmatrix}$

$$\begin{pmatrix} -\frac{12}{5} & \frac{4}{5} \\ \frac{4}{5} & -\frac{4}{15} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -\frac{12}{5}x + \frac{4}{5}y = 0 \text{ so } y = 3x \therefore \mathbf{v}_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\therefore \lambda_1 = -2, \mathbf{v}_1 = \begin{pmatrix} -3 \\ 1 \end{pmatrix} \text{ and } \lambda_2 = \frac{2}{3}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

6)

$$\begin{pmatrix} 6 & -2 \\ k-3 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 6 & -2 \\ k-3 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 6-\lambda & -2 \\ k-3 & 2-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (6-\lambda)(2-\lambda) + 2(k-3) = \lambda^2 - 8\lambda + 12 + 2(k-3) = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 6 + 2k = 0$$

Discriminant: $b^2 - 4ac = 0$

$$\therefore (-8)^2 - 4(6+2k) = 0$$

$$\Rightarrow 64 - 24 - 8k = 0$$

$$\therefore k = 5$$

7 a)

$$\begin{pmatrix} 4 & 1 \\ k & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 4 & 1 \\ k & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4-\lambda & 1 \\ k & 3-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (4-\lambda)(3-\lambda) - k = \lambda^2 - 7\lambda + 12 - k = 0$$

Discriminant: $b^2 - 4ac > 0$

$$\therefore (-7)^2 - 4(12-k) > 0$$
$$\Rightarrow 49 - 48 + 4k = 0$$
$$\therefore k > -\frac{1}{4}$$

7 b)

$$\begin{pmatrix} k & 2 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} k & 2 \\ 3 & 5 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} k-\lambda & 2 \\ 3 & 5-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (k-\lambda)(5-\lambda) - 6 = \lambda^2 - (k+5)\lambda + (5k-6) = 0$$

Discriminant: $b^2 - 4ac > 0$

$$\therefore (-(k+5))^2 - 4(5k-6) > 0$$
$$\Rightarrow k^2 - 10k + 49 > 0$$
$$\Rightarrow (k-5)^2 + 24 > 0$$

Since $(k-5)^2 \geq 0$ and $24 > 0$

$$\therefore k \in \mathbb{R}$$

8 a)

$$\begin{pmatrix} -1 & k \\ 4 & k-1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} -1 & k \\ 4 & k-1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1-\lambda & k \\ 4 & k-1-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (-1-\lambda)(k-1-\lambda) - 4k = \lambda^2 + (2-k)\lambda - k + 1 - 4k$$

$$\Rightarrow \lambda^2 + (2-k)\lambda + 1 - 5k = 0$$
$$\therefore \lambda^2 + (2-k)\lambda - (5k-1) = 0 \quad \text{as req.}$$

8 b)

Discriminant: $b^2 - 4ac > 0$

$$\therefore (2-k)^2 - 4(1-5k) > 0$$
$$\Rightarrow k^2 - 4k + 4 - 4 + 20k > 0 \Rightarrow k^2 + 16k > 0$$
$$\Rightarrow k(k+16) > 0$$
$$\therefore k < -16 \text{ or } k > 0$$

8 c)

When $k = 9$

$$\lambda^2 - 7\lambda - 44 = 0$$
$$\therefore \lambda = -4 \text{ or } \lambda = 11$$

When $\lambda_1 = -4$ then $\mathbf{A} + 4\mathbf{I} = \begin{pmatrix} 3 & 9 \\ 4 & 12 \end{pmatrix}$

$$\begin{pmatrix} 3 & 9 \\ 4 & 12 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow 3x + 9y = 0 \text{ so } 3y = -x \therefore \mathbf{v}_1 = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

When $\lambda_2 = 11$ then $\mathbf{A} - 11\mathbf{I} = \begin{pmatrix} -12 & 9 \\ 4 & -3 \end{pmatrix}$

$$\begin{pmatrix} -12 & 9 \\ 4 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -12x + 9y = 0 \text{ so } 3y = 4x \therefore \mathbf{v}_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$\therefore \lambda_1 = -4, \mathbf{v}_1 = \begin{pmatrix} -3 \\ 1 \end{pmatrix} \text{ and } \lambda_2 = 11, \mathbf{v}_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

9 a)

$$\begin{pmatrix} p & -q \\ q & p \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$
$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} p & -q \\ q & p \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p-\lambda & -q \\ q & p-\lambda \end{pmatrix}$$
$$\det(\mathbf{A} - \lambda\mathbf{I}) = (p-\lambda)(p-\lambda) + q^2 = \lambda^2 - 2p\lambda + p^2 + q^2 = 0$$

Discriminant: $b^2 - 4ac$

$$\therefore (-2p)^2 - 4(p^2 + q^2) = 4p^2 - 4p^2 - 4q^2 = -4q^2$$
$$\therefore \lambda = \frac{2p \pm \sqrt{-4q^2}}{2} = \frac{2p \pm 2qi}{2} = p \pm qi$$

9 b)

$$\begin{pmatrix} p & q \\ 0 & p \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$
$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} p & q \\ 0 & p \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p-\lambda & q \\ 0 & p-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (p - \lambda)(p - \lambda) = \lambda^2 - 2p\lambda + p^2 = 0$$

Discriminant: $b^2 - 4ac$

$$\therefore (-2p)^2 - 4(p^2) = 4p^2 - 4p^2 = 0$$

$$\therefore \lambda = \frac{2p \pm \sqrt{0}}{2} = \frac{2p}{2} = p$$

10)

If λ is an eigenvalue of $\mathbf{M}$, then there exists a non-zero vector $\mathbf{v}$ such that $\mathbf{Mv} = \lambda\mathbf{v}$

Prove true for $n = 1$:

$\mathbf{M}^1\mathbf{v} = \mathbf{Mv} = \lambda\mathbf{v} = \lambda^1\mathbf{v}$ so true for $n = 1$

Assume true for $n = k$:

$$\mathbf{M}^k\mathbf{v} = \lambda^k\mathbf{v}$$

Prove true for $n = k + 1$:

$$\begin{aligned}\mathbf{M}^{k+1}\mathbf{v} &= \mathbf{M}^1(\mathbf{M}^k\mathbf{v}) \\ &= \mathbf{M}(\lambda^k\mathbf{v}) \\ &= \lambda^k\mathbf{Mv} \\ &= \lambda^k\lambda\mathbf{v} \\ &= \lambda^{k+1}\mathbf{v}\end{aligned}$$

Which is $\mathbf{M}^n\mathbf{v} = \lambda^n\mathbf{v}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

11 a)

$3 \times 2 = 6$

11 b)

$4^2 \times 5 = 80$

11 c)

$-1 \times 2^3 = -8$

11 d)

$(-2)^2 \times (-4)^3 = -256$

11 e)

$p^m q^n$